Gauss-Seidel Iteration Method

We use the updated values of $x_1, x_2, \ldots, x_{i-1}$ in computing the value of the variable x_i. We assume that the pivots $a_{ii} \neq 0$, for all i. We write the equations as

$$a_{11}x_1 = b_1 - (a_{12}x_2 + a_{13}x_3)$$
$$a_{22}x_2 = b_2 - (a_{21}x_1 + a_{23}x_3)$$
$$a_{33}x_3 = b_3 - (a_{31}x_1 + a_{32}x_2)$$

The Gauss-Seidel iteration method is defined as

$$x_1^{(k+1)} = \frac{1}{a_{11}}[b_1 - (a_{12}x_2^{(k)} + a_{13}x_3^{(k)})]$$

$$x_2^{(k+1)} = \frac{1}{a_{22}}[b_2 - (a_{21}x_1^{(k+1)} + a_{23}x_3^{(k)})] \qquad (1)$$

$$x_3^{(k+1)} = \frac{1}{a_{33}}[b_3 - (a_{31}x_1^{(k+1)} + a_{32}x_2^{(k+1)})]$$

$$k = 0, 1, 2,...$$

This method is also called the *method of successive displacement.*

We observe that (1) is same as writing the given system as

$$\begin{aligned} a_{11}x_1^{(k+1)} &= b_1 - (a_{12}x_2^{(k)} + a_{13}x_3^{(k)}) \\ a_{21}x_1^{(k+1)} + a_{22}x_2^{(k+1)} &= b_2 - a_{23}x_3^{(k)} \\ a_{31}x_1^{(k+1)} + a_{32}x_2^{(k+1)} + a_{33}x_3^{(k+1)} &= b_3 \end{aligned} \qquad (2)$$

Remark *A sufficient condition for convergence of the Gauss-Seidel method is that the system of equations is diagonally dominant*, that is, the coefficient matrix **A** is diagonally dominant. This implies that convergence may be obtained even if the system is not diagonally dominant. If the system is not diagonally dominant, we may exchange the equations, if possible, such that the new system is diagonally dominant and convergence is guaranteed. The necessary and sufficient condition for convergence is that the spectral radius of the iteration matrix **H** is less than one unit, that is, $\rho(\mathbf{H}) < 1$, where $\rho(\mathbf{H})$ is the largest eigen value in magnitude of **H.** Testing of this condition is beyond the scope of the syllabus.

If both the Gauss-Jacobi and Gauss-Seidel methods converge, then *Gauss-Seidel method converges at least two times faster than the Gauss-Jacobi method.*

Example *Find the solution of the system of equations*

$$45x_1 + 2x_2 + 3x_3 = 58$$

$$-3x_1 + 22x_2 + 2x_3 = 47$$

$$5x_1 + x_2 + 20x_3 = 67$$

correct to three decimal places, using the Gauss-Seidel iteration method.

Solution The given system of equations is strongly diagonally dominant. Hence, we can expect fast convergence. Gauss-Seidel method gives the iteration

$$x_1^{(k+1)} = \frac{1}{45}\,(58 - 2x_2^{(k)} - 3x_3^{(k)}),$$

$$x_2^{(k+1)} = \frac{1}{22}\,(47 + 3x_1^{(k+1)} - 2x_3^{(k)}),$$

$$x_3^{(k+1)} = \frac{1}{20}\,(67 - 5x_1^{(k+1)} - x_2^{(k+1)}).$$

Starting with $x_1^{(0)} = 0$, $x_2^{(0)} = 0$, $x_3^{(0)} = 0$, we get the following results.

First iteration

$$x_1^{(1)} = \frac{1}{45}\,(58 - 2x_2^{(0)} - 3x_3^{(0)}) = \frac{1}{45}\,(58) = 1.28889,$$

$$x_2^{(1)} = \frac{1}{22}\,(47 + 3x_1^{(1)} - 2x_3^{(0)}) = \frac{1}{22}\,(47 + 3(1.28889) - 2(0)) = 2.31212,$$

$$x_3^{(1)} = \frac{1}{20}\,(67 - 5x_1^{(1)} - x_2^{(1)}) = \frac{1}{20}\,(67 - 5(1.28889) - (2.31212)) = 2.91217.$$

Second iteration

$$x_1^{(2)} = \frac{1}{45}\,(58 - 2x_2^{(1)} - 3x_3^{(1)}) = \frac{1}{45}\,(58 - 2(2.31212) - 3(2.91217)) = 0.99198,$$

$$x_2^{(2)} = \frac{1}{22}\,(47 + 3x_1^{(2)} - 2x_3^{(1)}) = \frac{1}{22}\,(47 + 3(0.99198) - 2(2.91217)) = 2.00689,$$

$$x_3^{(2)} = \frac{1}{20}\,(67 - 5x_1^{(2)} - x_2^{(2)}) = \frac{1}{20}\,(67 - 5(0.99198) - (2.00689)) = 3.00166.$$

Third iteration

$$x_1^{(3)} = \frac{1}{45}\,(58 - 2x_2^{(2)} - 3x_3^{(2)}) = \frac{1}{45}\,(58 - 2(2.00689) - 3(3.00166) = 0.99958,$$

$$x_2^{(3)} = \frac{1}{22}\,(47 + 3x_1^{(3)} - 2x_3^{(2)}) = \frac{1}{22}\,(47 + 3(0.99958) - 2(3.00166)) = 1.99979,$$

$$x_3^{(3)} = \frac{1}{20}\,(67 - 5x_1^{(3)} - x_2^{(3)}) = \frac{1}{20}\,(67 - 5(0.99958) - (1.99979)) = 3.00012.$$

Fourth iteration

$$x_1^{(4)} = \frac{1}{45}(58 - 2x_2^{(3)} - 3x_3^{(3)}) = \frac{1}{45}(58 - 2(1.99979) - 3(3.00012)) = 1.00000,$$

$$x_2^{(4)} = \frac{1}{22}(47 + 3x_1^{(4)} - 2x_3^{(3)}) = \frac{1}{22}(47 + 3(1.00000) - 2(3.00012)) = 1.99999,$$

$$x_3^{(4)} = \frac{1}{20}(67 - 5x_1^{(4)} - x_2^{(4)}) = \frac{1}{20}(67 - 5(1.00000) - (1.99999)) = 3.00000.$$

We find $\left| x_1^{(4)} - x_1^{(3)} \right| = | \ 1.00000 - 0.99958 \ | = 0.00042,$

$\left| x_2^{(4)} - x_2^{(3)} \right| = | \ 1.99999 - 1.99979 \ | = 0.00020,$

$\left| x_3^{(4)} - x_3^{(3)} \right| = | \ 3.00000 - 3.00012 \ | = 0.00012.$

Since, all the errors in magnitude are less than 0.0005, the required solution is

$x_1 = 1.0, \ x_2 = 1.99999, \ x_3 = 3.0.$

Example *Computationally show that Gauss-Seidel method applied to the system of equations*

$$3x_1 - 6x_2 + 2x_3 = 23$$
$$-4x_1 + x_2 - x_3 = -8$$
$$x_1 - 3x_2 + 7x_3 = 17$$

diverges. Take the initial approximations as $x_1 = 0.9$, $x_2 = -3.1$, $x_3 = 0.9$. *Interchange the first and second equations and solve the resulting system by the Gauss-Seidel method. Again take the initial approximations as* $x_1 = 0.9$, $x_2 = -3.1$, $x_3 = 0.9$, *and obtain the result correct to two decimal places. The exact solution is* $x_1 = 1.0$, $x_2 = -3.0$, $x_3 = 1.0$.

Solution Note that the system of equations is not diagonally dominant. Gauss-Seidel method gives the iteration

$$x_1^{(k+1)} = [23 + 6x_2^{(k)} - 2x_3^{(k)})]/3$$
$$x_2^{(k+1)} = [-8 + 4x_1^{(k+1)} + x_3^{(k)}]$$
$$x_3^{(k+1)} = [17 - x_1^{(k+1)} + 3x_2^{(k+1)}]/7.$$

Starting with the initial approximations $x_1 = 0.9$, $x_2 = -3.1$, $x_3 = 0.9$, we obtain the following results.

First iteration

$$x_1^{(1)} = \frac{1}{3}\,[23 + 6x_2^{(0)} - 2x_3^{(0)}] = \frac{1}{3}\,[23 + 6(-3.1) - 2(0.9)] = 0.8667,$$

$$x_2^{(1)} = [-8 + 4x_1^{(1)} + x_3^{(0)}] = [-8 + 4(0.8667) + 0.9] = -3.6332,$$

$$x_3^{(1)} = \frac{1}{7}\,[17 - x_1^{(1)} + 3x_2^{(1)}] = \frac{1}{7}\,[17 - (0.8667) + 3(-3.6332)] = 0.7477.$$

Second iteration

$$x_1^{(2)} = \frac{1}{3}\,[23 + 6x_2^{(1)} - 2x_3^{(1)}] = \frac{1}{3}\,[23 + 6\,(-3.6332) - 2(0.7477)] = -0.0982,$$

$$x_2^{(2)} = [-8 + 4x_1^{(2)} + x_3^{(1)}] = [-8 + 4(-0.0982) + 0.7477] = -7.6451,$$

$$x_3^{(2)} = \frac{1}{7}\,[17 - x_1^{(2)} + 3x_2^{(2)}] = \frac{1}{7}\,[17 + 0.0982 + 3(-7.6451)] = -0.8339.$$

Third iteration

$$x_1^{(3)} = \frac{1}{3}\,[23 + 6x_2^{(2)} - 2x_3^{(2)}] = \frac{1}{3}\,[23 + 6\,(-7.6451) - 2(-0.8339)] = -7.0676,$$

$$x_2^{(3)} = [-8 + 4x_1^{(3)} + x_3^{(2)}] = [-8 + 4(-7.0676) - 0.8339] = -37.1043,$$

$$x_3^{(3)} = \frac{1}{7}\,[17 - x_1^{(3)} + 3x_2^{(3)}] = \frac{1}{7}\,[17 + 7.0676 + 3(-37.1043)] = -12.4636.$$

It can be observed that the iterations are diverging very fast.

Now, we exchange the first and second equations to obtain the system

$$-4x_1 + x_2 - x_3 = -8$$
$$3x_1 - 6x_2 + 2x_3 = 23$$
$$x_1 - 3x_2 + 7x_3 = 17.$$

The system of equations is now diagonally dominant. Gauss-Seidel method gives iteration

$$x_1^{(k+1)} = [8 + x_2^{(k)} - x_3^{(k)}]/4$$
$$x_2^{(k+1)} = -[23 - 3x_1^{(k+1)} - 2x_3^{(k)}]/6$$
$$x_3^{(k+1)} = [17 - x_1^{(k+1)} + 3x_2^{(k+1)}]/7.$$

Starting with the initial approximations $x_1 = 0.9$, $x_2 = -3.1$, $x_3 = 0.9$, we obtain the following results.

First iteration

$$x_1^{(1)} = \frac{1}{4}[8 + x_2^{(0)} - x_3^{(0)}] = \frac{1}{4}[8 - 3.1 - 0.9] = 1.0,$$
$$x_2^{(1)} = -\frac{1}{6}[23 - 3x_1^{(1)} - 2x_3^{(0)}] = -\frac{1}{6}[23 - 3(1.0) - 2(0.9)] = -3.0333,$$
$$x_3^{(1)} = \frac{1}{7}[17 - x_1^{(1)} + 3x_2^{(1)}] = \frac{1}{7}[17 - 1.0 + 3(-3.0333)] = 0.9857.$$

Second iteration

$$x_1^{(2)} = \frac{1}{4}\,[8 + x_2^{(1)} - x_3^{(1)}] = \frac{1}{4}\,[8 - 3.0333 - 0.9857] = 0.9953,$$

$$x_2^{(2)} = -\frac{1}{6}\,[23 - 3x_1^{(2)} - 2x_3^{(1)}] = -\frac{1}{6}\,[23 - 3(0.9953) - 2(0.9857)] = -3.0071,$$

$$x_3^{(2)} = \frac{1}{7}\,[17 - x_1^{(2)} + 3x_2^{(2)}] = \frac{1}{7}\,[17 - 0.9953 + 3(-3.0071)] = 0.9976.$$

Third iteration

$$x_1^{(3)} = \frac{1}{4}\,[8 + x_2^{(2)} - x_3^{(2)}] = \frac{1}{4}\,[8 - 3.0071 - 0.9976] = 0.9988,$$

$$x_2^{(3)} = -\frac{1}{6}\,[23 - 3x_1^{(3)} - 2x_3^{(2)}] = -\frac{1}{6}\,[23 - 3(0.9988) - 2(0.9976)] = -3.0014,$$

$$x_3^{(3)} = \frac{1}{7}\,[17 - x_1^{(3)} + 3x_2^{(3)}] = \frac{1}{7}\,[17 - 0.9988 + 3(-3.0014)] = 0.9996.$$

Fourth iteration

$$x_1^{(4)} = \frac{1}{4}\,[8 + x_2^{(3)} - x_3^{(3)}] = \frac{1}{4}\,[8 - 3.0014 - 0.9996] = 0.9998,$$

$$x_2^{(4)} = -\frac{1}{6}\,[23 - 3x_1^{(4)} - 2x_3^{(3)}] = -\frac{1}{6}\,[23 - 3(0.9998) - 2(0.9996)] = -3.0002,$$

$$x_3^{(4)} = \frac{1}{7}\,[17 - x_1^{(4)} + 3x_2^{(4)}] = \frac{1}{7}\,[17 - 0.9998 + 3(-3.0002)] = 0.9999.$$

We find $\left| x_1^{(4)} - x_1^{(3)} \right| = |\ 0.9998 - 0.9988\ | = 0.0010,$

$$\left| x_2^{(4)} - x_2^{(3)} \right| = |\ -3.0002 + 3.0014\ | = 0.0012,$$

$$\left| x_3^{(4)} - x_3^{(3)} \right| = |\ 0.9999 - 0.9996\ | = 0.0003.$$

Since, all the errors in magnitude are less than 0.005, the required solution is

$$x_1 = 0.9998,\ x_2 = -3.0002,\ x_3 = 0.9999.$$